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Conceptual Framework for Deepfake Audio Detection in Call Centers Using Spectral and Temporal Analysis

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Chapter 1

INTRODUCTION

INTRODUCTION

1.1 Research Background

The rapid evolution of deep learning techniques has given rise to new and complex challenges in the manipulation of digital media. Image, audio, and video files are increasingly being manipulated using artificial intelligence-based methods known as “deepfake”, a portmanteau of deep learning and fake [1][2]. The term “deepfake” first entered the public domain in 2017, when a Reddit user with the username “deepfakes” shared manipulated videos online. These videos were created using Google’s open-source deep learning technology to swap faces in existing footage, demonstrating the potential of AI-driven synthetic media [3].

Between 2018 and 2019, there was a significant increase in the number of articles about deepfake (from 60 to 309) [4]. Articles about deepfakes were expected to increase to over 730 by 2020’s end, according to predictions made on July 24 [5]. In [6] found that most focus is on video deepfakes, particularly in developing video deepfakes.

Deepfake refers to a form of digitally created content in which artificial intelligence techniques are used to manipulate or generate audio, video, or images, often by replacing or imitating the appearance or voice of real individuals. Among these applications, the use of deepfake techniques in audio is particularly concerning, as it enables the creation and distribution of highly realistic but fake speech recordings. This raises serious concerns regarding the reliability, authenticity, and security of audio-based communication systems [7].

There are several approaches to creating deepfake audio, with the three most commonly used techniques being text-to-speech (TTS), voice conversion (VC), and replay attacks [7].

Text to Speech TTS (speech synthesis (SS))

Text to speech is a technique that generates artificial speech directly from textual input using deep learning models. Modern TTS systems are capable of producing highly natural and human-like voices, often mimicking specific individuals by learning their vocal characteristics. These systems typically rely on neural architectures such as Tacotron and WaveNet, which model both linguistic and acoustic features to synthesize realistic speech. In malicious contexts, TTS can be used to create convincing fake audio without requiring the attacker to speak [7][8][9].

Voice Conversion VC

Voice conversion involves transforming the voice of a source speaker to sound like that of a target speaker while preserving the original linguistic content. Unlike text to speech, which generates speech from text, voice conversion modifies existing speech by altering acoustic features such as

pitch, timbre, and speaking style. This technique is particularly dangerous in real time scenarios, as attackers can speak naturally while their voice is converted to impersonate another individual during live interactions [7].

Replay Attacks

Replay attacks involve the use of pre-recorded genuine speech to impersonate a legitimate speaker. In this approach, an attacker records a target individual's voice and replays it to bypass authentication systems. Unlike text to speech and voice conversion, replay attacks do not involve synthetic generation or transformation of speech but instead rely on authentic recordings. However, they may still include channel distortions such as background noise or compression artifacts introduced during recording and playback [10][11][12].

The rapid advancement of generative artificial intelligence has created an unprecedented challenge for voice-based authentication systems, particularly in high stakes call center environments. Voice authentication has become a critical biometric method for financial institutions, telecommunications providers, and customer service operations, supporting millions of daily interactions that require reliable identity verification [13]. However, the emergence of advanced text to speech and voice conversion technologies has exposed traditional voice biometric systems to synthetic speech attacks that can convincingly imitate legitimate users.

Call centers have become attractive targets for deepfake enabled fraud due to their reliance on audio-based authentication processes. Modern attackers employ multi vector strategies that combine phishing, credential harvesting, and AI driven voice synthesis to carry out account takeovers and unauthorized transactions. The financial implications are significant, with a large proportion of account takeover incidents in financial services originating from call center interactions. Traditional voice biometric systems analyze vocal characteristics such as pitch patterns, speaking rhythm, and frequency distribution to generate voiceprints, yet these systems lack mechanisms to verify whether the speech itself is genuine or artificially generated.

In response to these threats, deepfake detection methods have evolved toward architectures that analyze both spectral and temporal characteristics of speech signals. Spectro temporal features, including Linear Frequency Cepstral Coefficients, Mel Frequency Cepstral Coefficients, Constant Q Cepstral Coefficients, and log power spectrograms, capture frequency-based patterns and short-term variations that reveal artifacts introduced during speech synthesis. At the same time, temporal analysis focuses on speech dynamics such as pause patterns, rhythm consistency, and prosodic variations, which often expose unnatural patterns in synthetic audio. The integration of these complementary approaches has demonstrated improved performance in detecting previously unseen spoofing attacks.

Recent advances in self-supervised learning have further enhanced deepfake audio detection capabilities. Models such as WavLM, Wav2Vec 2.0, and XLS R learn robust speech representations from large scale unlabeled data. In particular, WavLM incorporates speech denoising and prediction tasks, enabling it to capture both speaker specific attributes and environmental characteristics that are useful for detecting synthetic artifacts. When combined with efficient classification models, these approaches achieve high detection accuracy while maintaining computational efficiency suitable for real time applications.

Despite these advancements, call center environments impose strict operational constraints on detection systems. Real time voice authentication requires processing within a few seconds without interrupting natural conversation flow, demanding a balance between detection accuracy and computational efficiency. In addition, telephony conditions such as limited sampling rates, compression artifacts, and transmission noise introduce distortions that can obscure synthetic speech characteristics. These challenges highlight the need for robust and efficient detection frameworks tailored specifically for real time call center deployment.

1.2 Problem Statement

Despite significant advances in deepfake audio detection, a critical gap persists between research-oriented systems and deployable call center solutions. Existing detection frameworks are predominantly developed as standalone laboratory models, optimized for clean and high-fidelity audio conditions, with limited consideration for real-world telephony environments. As a result, they lack effective integration with telephony infrastructure and real-time authentication workflows. This disconnect creates a vulnerability window in which sophisticated synthetic speech attacks can bypass security mechanisms before detection occurs.

The key challenges addressed in this research include:

Generalization Deficit:

Current detection models often experience significant performance degradation when exposed to unseen synthesis techniques or cross-domain conditions, particularly those involving telephony distortions. The rapid advancement of text-to-speech (TTS) and voice conversion (VC) technologies necessitates the development of adaptive detection systems capable of identifying emerging attack methods without requiring frequent retraining.

Real-time Processing Constraints:

Call center authentication systems require near-instantaneous decision-making within live conversational settings. However, many high-performing detection models rely on computationally intensive architectures that fail to meet strict latency requirements. This creates a

critical trade-off between detection accuracy and processing efficiency, limiting their practical deployment.

Environmental Robustness:

Telephony channels introduce various distortions, including compression, packet loss, and background noise, which can obscure the distinguishing features of synthetic speech. These conditions often result in overlapping feature distributions between genuine and spoofed audio, making accurate classification more challenging.

Integration Complexity:

Most existing detection systems lack standardized interfaces, such as APIs or microservices architectures, required for seamless integration with operational platforms like interactive voice response (IVR) systems, customer relationship management (CRM) tools, and fraud detection pipelines. This limits their usability in real-world applications.

In light of these challenges, the fundamental problem addressed by this research is the lack of a unified conceptual framework that integrates spectral and temporal analysis techniques within a real-time, call center-oriented architecture. Such a framework must support adaptive learning, robust performance under telephony conditions, and seamless integration into existing operational systems.

1.3 Research Questions

This study seeks to address the following research questions:

RQ1.What are the limitations of existing deepfake audio detection approaches in call center environments?

RQ2How can spectral and temporal analysis techniques be effectively combined within a conceptual framework for detecting deepfake audio?

RQ3.What key features (spectral and temporal) are most suitable for identifying voice spoofing attacks in telephony-based systems?

RQ4.How can a conceptual framework be designed to support real-time detection of deepfake audio in call center environments?

RQ5.How can the proposed framework be structured to ensure robustness against telephony channel distortions and emerging deepfake techniques?

1.4 Research Aims and Objectives

Overall Purpose of the Study

The overall purpose of this study is to develop a conceptual framework for detecting deepfake audio in call center environments using spectral and temporal analysis techniques, ensuring robustness, real-time capability, and operational feasibility.

Aims

- 1.To design a conceptual detection framework that leverages spectral and temporal features to identify deepfake audio in call center communications.
- 2.To propose an operational integration model that enables real-time deployment within existing call center infrastructures.

Objectives

- 1.To **review** existing deepfake audio detection techniques and identify their limitations in call center environments.
2. To **examine** spectral and temporal feature extraction methods relevant to detecting synthetic speech.
- 3.To **design** a conceptual framework that integrates spectral and temporal analysis for deepfake audio detection.
4. To **define** the key components and processes involved in the proposed detection framework.
- 5.To **evaluate** how the proposed framework can support real-time detection in telephony-based systems.
- 6.To **propose** strategies for ensuring robustness and adaptability against evolving deepfake audio techniques.

1.5 Research Timeline

The research will be carried out over an approximate three-month period, spanning from January to late March 2026. This timeline is designed to ensure a structured and efficient progression of all research activities. The detailed schedule is presented below:

Task Name	Start Date	Finish Date	Duration
Research	01 Jan 2026	10 Jan 2026	10 Days
Chapter Two: Literature Review	11 Jan 2026	31 Jan 2026	20 Days
Chapter One: Introduction	01 Feb 2026	10 Feb 2026	10 Days
Chapter Three: Methodology	11 Feb 2026	10 March 2026	28 Days
Revision	11 March 2026	18 March 2026	08 Days
Completion	19 March 2026	26 March 2026	08 Days
Submission	27 March 2026	29 March 2026	03 Days

Table 1.1: Research Timeline.

In addition, the research timeline is presented using a Gantt chart to provide a clear visual representation of the sequence, duration, and overlap of the various research activities.

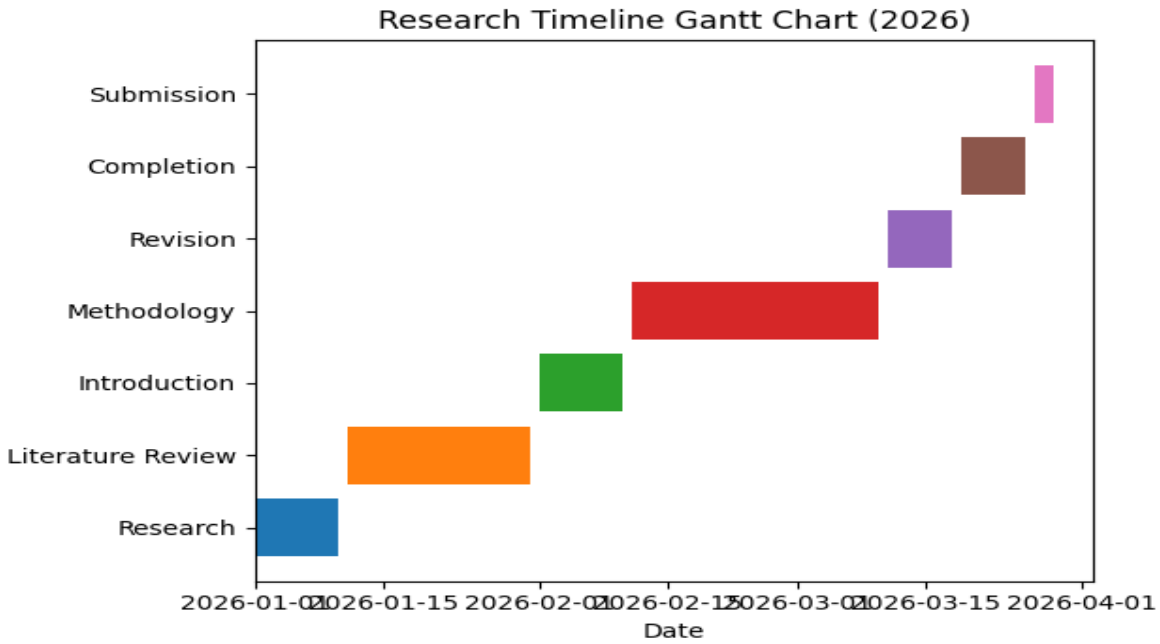


Figure 1.1: Research Timeline Gantt Chart

1.6 Research Organization

This research is organized into four main chapters:

- **Chapter 1: Introduction** – Presents the research background, problem statement, aims, objectives, and structure.
- **Chapter 2: Literature Review** – Reviews existing deepfake detection techniques, spectral and temporal analysis methods.
- **Chapter 3: Research Methodology and Proposed Conceptual Framework** – This chapter presents the methodology adopted in the study, including the research design, feature extraction techniques, and evaluation strategy. It further introduces the proposed conceptual framework for deepfake audio detection, detailing the architecture, processing pipeline, and integration strategy for deployment in call center environments.
- **Chapter 4: Discussion and Conclusion** – Summarizes findings, discusses implications, limitations, and suggests directions for future research.

Chapter 2

LITERATURE REVIEW

LITERATURE REVIEW

This section reviews existing studies on deepfake audio and voice spoofing detection, with a focus on techniques based on spectral and temporal analysis. Several machine learning and deep learning approaches have been proposed to detect spoofed speech, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), and hybrid architectures. The following studies highlight key advancements, performance metrics, and limitations in this domain.

Shahzad et al. (2025) introduced a hybrid VGGish-LSTM model to address voice spoofing attacks targeting Automatic Speaker Verification (ASV) systems. The approach utilizes transfer learning-based embeddings extracted via the VGGish model to capture both spectral and temporal features, which are then processed by an LSTM network to model long-term dependencies in speech. Evaluated on the ASVspoof 2019 Logical Access dataset, the model achieved an accuracy of 99.11% and an EER of 1.18%, outperforming CNN-based and YAMNet-based alternatives. Despite its strong performance, the study highlighted limitations such as dataset dependency, potential overfitting, and limited coverage of emerging deepfake attack types [14].

Verma et al. (2025) conducted a comparative study of various deep learning models, including Bidirectional Long Short-Term Memory (BLSTM) and Convolutional Neural Networks (CNNs), for deepfake audio detection. The models were evaluated using both time-domain features, such as Zero Crossing Rate (ZCR) and Root Mean Square Energy (RMSE), and spectral features such as MFCCs extracted from the ASVspoof 2019 dataset. The best-performing model, a Squeeze-and-Excitation Enhanced 1D CNN, achieved 97.64% accuracy, along with high precision, recall, and F1-score values of approximately 98%. However, the study noted several limitations, including dataset dependency, lack of diversity in accents and speech styles, limited evaluation of complex attack types, and high computational cost, which may hinder real-world deployment [15].

Shah (2025) proposed a comprehensive AI-driven framework for fraud detection across mobile communication channels, including SMS, phone calls, and messaging platforms. The study employed deep learning architectures such as Long Short-Term Memory (LSTM) and hybrid CNN-LSTM models, alongside acoustic features like Mel-Frequency Cepstral Coefficients (MFCCs) and spectral analysis to detect AI-generated voice scams. The model achieved high accuracy levels ranging from 94% to 99% across different tasks. The study also emphasized the importance of multi-layered detection systems that integrate real-time anomaly detection with behavioral analysis to address increasingly sophisticated deepfake-based attacks [16].

Kwak et al. (2020) proposed a lightweight voice spoofing detection architecture known as ResMax, which combines the skip connection concept of residual networks (ResNet) with Max

Feature Map (MFM) activation from Light CNN. The model utilizes Constant Q Transform (CQT) spectral features to extract frequency-domain characteristics that distinguish between genuine and spoofed speech. Experimental results on the ASVspoof 2019 dataset show that the model achieved an Equal Error Rate (EER) of 0.37% on the Physical Access (PA) dataset and 2.19% on the Logical Access (LA) dataset, outperforming even ensemble systems on PA data. Furthermore, the model's low complexity, with only 286K parameters, demonstrates its suitability for real-time and resource-constrained environments [17].

Chapter 3

METHODOLOGY

3.1 Introduction

This chapter presents a comprehensive description of the methodology adopted in this research project titled *A Conceptual Framework for Deepfake Audio Detection in Call Center Environments Using Spectral and Temporal Analysis*, outlining the design, components, and processes of the proposed framework.

3.2 Conceptual Framework Overview

The proposed framework adopts a multi stage pipeline architecture designed for integration with call center interactive voice response systems. The framework processes incoming audio streams in overlapping windows of 3 to 5 seconds to balance detection latency with temporal context requirements.

As illustrated in Figure 2, the framework consists of four main modules:

1. Telecommunication preprocessing and normalization
2. Temporal spectral hybrid feature extraction
3. Dual stream analysis engine
4. Decision fusion and risk scoring

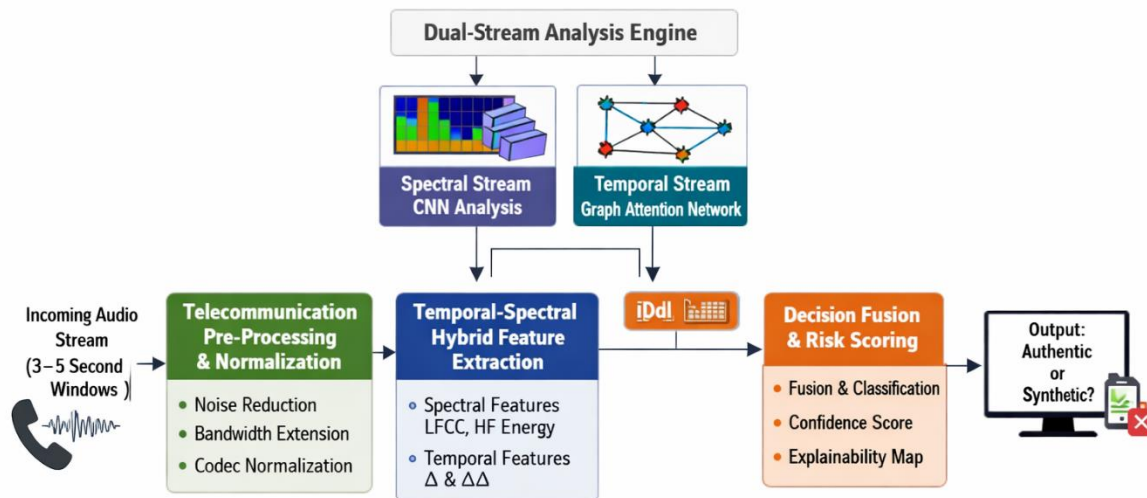


Figure 2. The proposed conceptual framework for deepfake audio detection in call center environments

3.2 Telecommunication Preprocessing Module

Given the prevalence of codec compression in call center infrastructures, this module focuses on preparing audio signals for reliable analysis. Bandwidth extension techniques are applied to reconstruct high frequency components in the 4 to 8 kHz range, which may contain important indicators of synthetic speech.

In addition, adaptive noise reduction methods are applied to handle typical call center noise sources such as background conversations, keyboard typing, and environmental noise, while preserving subtle artifacts associated with synthetic audio.

The module also performs codec aware normalization by identifying compression standards such as G.711, GSM, and AMR through statistical analysis of quantization noise. Appropriate filtering techniques are then applied to standardize the spectral characteristics of the audio signal.

3.3 Temporal Spectral Hybrid Feature Extraction

The framework employs a temporal spectral hybrid feature extraction approach to capture both frequency- based and time- based characteristics of speech.

Spectral Features:

Linear Frequency Cepstral Coefficients are extracted using linearly spaced filters to preserve high frequency details that may reveal artifacts introduced by synthetic speech generation. Additional high frequency energy metrics are computed within the 3 to 8 kHz range to capture inconsistencies in energy distribution.

Temporal Features:

Temporal dynamics are captured using first and second order derivatives of spectral features, which reflect changes in speech over time. Statistical measures such as mean, variance, skewness, and kurtosis are computed across frames to describe temporal behavior. Genuine speech typically exhibits more variability compared to the smoother patterns observed in synthetic speech.

The resulting feature set provides a compact and interpretable representation suitable for analysis within the proposed framework.

3.4 Dual Stream Analysis Engine

The framework incorporates two parallel analysis streams designed to capture complementary characteristics of deepfake audio.

Spectral Stream:

A convolutional neural network- based approach is used to analyze spectral patterns and identify

anomalies in frequency structures, such as irregular harmonics and unnatural energy distributions.

Temporal Stream:

A graph- based attention mechanism is conceptually employed to model temporal relationships within the audio signal. This allows the system to identify inconsistencies in speech progression, rhythm, and transitions that may indicate synthetic generation.

The integration of these two streams enables a more comprehensive analysis by combining localized spectral patterns with broader temporal dependencies.

3.5 Decision Fusion and Continuous Learning

The outputs from the spectral and temporal streams are combined through a decision fusion mechanism. The framework assigns adaptive importance to each stream depending on the quality of the input signal. For example, in noisy conditions, greater emphasis may be placed on temporal consistency.

The final output includes:

- Classification of audio as authentic or synthetic
- Confidence score indicating the likelihood of spoofing
- Interpretability indicators highlighting key regions of interest

The framework also incorporates a conceptual continuous learning strategy, where newly identified attack patterns can be incorporated into the system to improve future detection performance.

3.6 Integration Architecture for Call Center Deployment

The framework is designed for integration within call center infrastructure, positioned between voice communication systems and speech processing modules.

Key design considerations include:

- Real time processing capability with minimal latency
- Compatibility with existing authentication and fraud detection systems
- Support for application programming interfaces to enable system integration
- Adaptability to varying network conditions such as packet loss and noise

3.7 Evaluation Methodology

The framework is evaluated using a multi- level strategy:

Benchmark Evaluation:

Standard datasets such as ASVspoof are used to assess detection performance under controlled conditions.

Telecommunication Simulation:

Audio signals are subjected to realistic telephony conditions including compression and noise to evaluate robustness.

Generalization Testing:

The framework is assessed against previously unseen attack types to evaluate adaptability.

Performance is measured using metrics such as Equal Error Rate, detection cost, processing time, and system efficiency.

Conclusion

This study presents a conceptual framework for deepfake audio detection in call center environments through the integration of spectral and temporal analysis techniques. By combining complementary feature representations and structured analysis components, the framework addresses key challenges related to real time processing, robustness, and adaptability. The proposed design provides a foundation for developing practical and efficient detection systems capable of operating in real world telephony environments.

References

- [1] Abbasi, Ahmed, et al. "A large-scale benchmark dataset for anomaly detection and rare event classification for audio forensics." *IEEE Access* 10 (2022): 38885-38894.
- [2] Wang, Tao, et al. "Prosody and voice factorization for few-shot speaker adaptation in the challenge m2voc 2021." *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021.
- [3] Javed, Abdul Rehman, et al. "A comprehensive survey on computer forensics: State-of-the-art, tools, techniques, challenges, and future directions." *IEEE Access* 10 (2022): 11065-11089.
- [4] Stupp, Catherine. "Fraudsters used AI to mimic CEO's voice in unusual cybercrime case." *The Wall Street Journal* 30.08 (2019).
- [5] Nguyen, Thanh Thi, et al. "Deep learning for deepfakes creation and detection: A survey." *Computer Vision and Image Understanding* 223 (2022): 103525.
- [6] Khanjani, Zahra, Gabrielle Watson, and Vandana P. Janeja. "How deep are the fakes? focusing on audio deepfake: A survey." *arXiv preprint arXiv:2111.14203* (2021).
- [7] Tahaoglu, Gul, et al. "Deepfake audio detection with spectral features and ResNeXt-based architecture." *Knowledge-Based Systems* 323 (2025): 113726.
- [8] Alshehri, Anfal, et al. "Audio deep fake detection with sonic sleuth model." *Computers* 13.10 (2024): 256.
- [9] Malinka, Kamil, et al. "Comprehensive multiparametric analysis of human deepfake speech recognition." *EURASIP Journal on Image and Video Processing* 2024.1 (2024): 24.
- [10] Kinnunen, Tomi, et al. "Asvspoof 2017: Automatic speaker verification spoofing and countermeasures challenge evaluation plan." *Training* 10.1508 (2017): 1508.
- [11] Kanwal, Tahira, et al. "Fake speech detection using VGGish with attention block." *EURASIP Journal on Audio, Speech, and Music Processing* 2024.1 (2024): 35.
- [12] Huang, Lian, and Chi-Man Pun. "Self-attention and hybrid features for replay and deep-fake audio detection." *arXiv preprint arXiv:2401.05614* (2024).
- [13] Safavi, Saeid, et al. "Fraud detection in voice-based identity authentication applications and services." *2016 IEEE 16th international conference on data mining workshops (ICDMW)*. IEEE, 2016.

[14] Shahzad, Komal, et al. "Enhancing voice spoofing detection: A hybrid approach with VGGish-LSTM model for improved security in automatic speaker verification systems." IEEE Access 13 (2025): 40682-40702.

[15] Verma, Kavya, et al. "Deepfake audio detection: A comparative study of advanced deep learning models." IEEE Access (2025).

[16] Shah, Kanchan D. "AI-DRIVEN FRAUD DETECTION AND PREVENTION FOR MOBILE COMMUNICATIONS: SMS, PHONE CALLS, AND WHATSAPP VIDEO CALLS." Journal of Science Research International (JSRI) ISSN 2456: 6365.

[17] Kwak, Il-Youp, et al. "ResMax: Detecting voice spoofing attacks with residual network and max feature map." 2020 25th International Conference on Pattern Recognition (ICPR). IEEE, 2021.